Generalizable Graph Foundation Models via Graphons and Size Shifts

1 Graphon operator setting, PE maps, and correspondence

Definition 1 (Graphon and graphon shift operator). *A (bounded) symmetric graphon is a measurable function $W : [0,1]^2 \to \mathbb{R}$ with $W(u,v) = W(v,u)$ and $\|W\|_\infty < \infty$. It induces a bounded self-adjoint integral operator $T_W : L^2([0,1]) \to L^2([0,1])$ given by*

$$(T_W f)(v) := \int_0^1 W(v,u) f(u)\, du.$$

Definition 2 (Induced step-graphon from a finite graph operator). *Fix $n \in \mathbb{N}$ and the partition $P_j = [(j-1)/n, j/n)$ for $j = 1, \ldots, n$. Given a finite (weighted) symmetric graph shift/operator $\Delta \in \mathbb{R}^{n\times n}$, define the induced self-adjoint step graphon (here $n\Delta$ equals adjacency matrix A)*

$$W_\Delta(u,v) := \sum_{i=1}^{n} \sum_{j=1}^{n} n\Delta_{ij}\, \mathbf{1}_{P_i}(u) \mathbf{1}_{P_j}(v),$$

and the induced operator T_{W_Δ}. For a node vector $x \in \mathbb{R}^n$, define the induced (step) graphon signal

$$(Sx)(u) := \sum_{j=1}^{n} x_j\, \mathbf{1}_{P_j}(u).$$

If π is a permutation of nodes, we write $\pi(\Delta)$ for the permuted operator and $W_{\pi(\Delta)}$ for the induced step graphon.

Definition 3 (Eigenvector positional encodings (Eig-PEs)). *Let T_W admit an eigendecomposition $\{(\lambda_\ell, \phi_\ell)\}_{\ell\geq 1}$ in $L^2([0,1])$, i.e. $T_W\phi_\ell = \lambda_\ell \phi_\ell$, $\|\phi_\ell\|_{L^2} = 1$. We index eigenpairs so that $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$. Fix $k \in \mathbb{N}$. Define the* graphon Coord-PE map

$$t_W^{eig} := (\phi_1, \ldots, \phi_k) \in L^2([0,1]; \mathbb{R}^k).$$

For a finite graph G equipped with graph shift operator Δ with eigenpairs $\{(\lambda_\ell, \varphi_\ell)\}_{\ell=1}^{n}$, , indexed so that $\lambda_1 \leq \cdots \leq \lambda_n$, where $\varphi_\ell \in \mathbb{R}^n$ are orthonormal under the scaled inner product $\langle x, y\rangle_n := \frac{1}{n} x^\top y$, define the graph Coord-PE tokens

$$t_G^{eig} := (\varphi_1, \ldots, \varphi_k) \in \mathbb{R}^{n\times k}$$

and the induced step *Eig-PE map on $[0,1]$,*

$$St_G^{eig} := (S\varphi_1, \ldots, S\varphi_k) \in L^2([0,1]; \mathbb{R}^k).$$

Definition 4 (Projector positional encodings (Proj-PEs))**.** *Let T_W admit an eigendecomposition $\{(\lambda_\ell, \phi_\ell)\}_{\ell \geq 1}$ in $L^2([0,1])$, i.e. $T_W \phi_\ell = \lambda_\ell \phi_\ell$ and $\|\phi_\ell\|_{L^2} = 1$. We index eigenpairs so that $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$. Fix $k \in \mathbb{N}$ and define the rank-k orthogonal projector*

$$\Pi_{W,k}: \ L^2([0,1]) \to L^2([0,1]), \qquad \Pi_{W,k} f := \sum_{\ell=1}^{k} \langle f, \phi_\ell \rangle_{L^2} \, \phi_\ell.$$

Equivalently, $\Pi_{W,k}$ has kernel

$$\Pi_{W,k}(u,v) := \sum_{\ell=1}^{k} \phi_\ell(u) \phi_\ell(v).$$

Define the graphon Proj-PE map with a m-dim readout function $r \in L^2([0,1]; \mathbb{R}^m)$, here $\Pi_{W,k}$ is extended to $L^2([0,1]; \mathbb{R}^m)$ channel-wise.

$$t_W^{\mathrm{proj}} \ := \ \Pi_{W,k} r \ \in \ L^2\big([0,1]; \, \mathbb{R}^m\big).$$

For a finite graph operator $\Delta \in \mathbb{R}^{n \times n}$ with eigenpairs $\{(\lambda_\ell, \varphi_\ell)\}_{\ell=1}^{n}$, indexed so that $\lambda_1 \leq \cdots \leq \lambda_n$, where $\varphi_\ell \in \mathbb{R}^n$ are orthonormal under the scaled inner product $\langle x, y \rangle_n := \frac{1}{n} x^\top y$, fix $k \leq n$ and denote

$$U_k = (\varphi_1, \ldots, \varphi_k) \in \mathbb{R}^{n \times k}, \qquad \Pi_k = \frac{1}{n} U_k U_k^\top \in \mathbb{R}^{n \times n}.$$

Define the graph Proj-PE tokens with a readout matrix $R = [r_1, \ldots, r_m] \in \mathbb{R}^{n \times m}$:

$$t_G^{\mathrm{proj}} \ := \ \Pi_k R \ \in \ \mathbb{R}^{n \times m}.$$

and the induced step Proj-PE map on $[0,1]$ with readout R by

$$St_G^{proj} := S(\Pi_k R) = (S(\Pi_k r_1), \ldots, S(\Pi_k r_m)) \in L^2([0,1]; \mathbb{R}^m)$$

For the induced step-graphon W_Δ, we take the default readout function as $r := SR$.

Proposition 1 (Correspondence for the induced step PE and induced graphon PE)**.** *Let $\Delta \in \mathbb{R}^{n \times n}$ and T_{W_Δ} be its induced graphon operator on $L^2([0,1])$. Let $\{(\lambda_j, \varphi_j)\}_{j=1}^{n}$ be an eigendecomposition of Δ. Then:*

1. *(**Eigenpair correspondence**) For every $j = 1, \ldots, n$,*

$$T_{W_\Delta} S\varphi_j = \lambda_j \, S\varphi_j.$$

2. *(**Eig-PE correspondence**) Fix $k \leq n$. For any $u \in P_i$,*

$$(St_G^{eig})(u) = t_{W_\Delta}^{eig}(u)$$

   provided the same ordering/sign convention is used for the corresponding eigenpairs.

3. *(**Projector-PE correspondence**) Fix $k \leq n$. For any $u \in P_i$,*

$$(St_G^{proj})(u) = t_{W_\Delta}^{proj}(u)$$

Proof. **Step-operator identity.** Fix any $x \in \mathbb{R}^n$ and let $v \in P_i$. Then $W_\Delta(v,u) = n\Delta_{ij}$ for $u \in P_j$, so

$$\begin{aligned}(T_{W_\Delta} Sx)(v) &= \int_0^1 W_\Delta(v,u)\,(Sx)(u)\,du = \sum_{j=1}^n \int_{P_j} n\Delta_{ij}\,x_j\,du \\ &= \sum_{j=1}^n n\Delta_{ij}\,x_j \cdot \lambda(P_j) = \sum_{j=1}^n n\Delta_{ij}\,x_j \cdot \frac{1}{n} = \sum_{j=1}^n \Delta_{ij}x_j = (\Delta x)_i.\end{aligned}$$

Since $(S(\Delta x))(v) = (\Delta x)_i$ for $v \in P_i$, we have the operator relation

$$T_{W_\Delta} S = S\Delta, \qquad \text{i.e.,} \qquad T_{W_\Delta}(Sx) = S(\Delta x) \ \ \forall x \in \mathbb{R}^n. \tag{1}$$

(1) Eigenpair correspondence. Let $\Delta\varphi_j = \lambda_j\varphi_j$. Applying (1) with $x = \varphi_j$ yields

$$T_{W_\Delta}(S\varphi_j) = S(\Delta\varphi_j) = S(\lambda_j\varphi_j) = \lambda_j\,S\varphi_j,$$

which proves (1).

(2) Eig-PE correspondence. Assume the discrete eigenvectors $\{\varphi_\ell\}_{\ell=1}^n$ are orthonormal under $\langle x,y\rangle_n = \frac{1}{n}x^\top y$ (equivalently $\|\varphi_\ell\|_2^2 = n$), so that $\|S\varphi_\ell\|_{L^2} = 1$ and $\{S\varphi_\ell\}$ is an L^2-orthonormal family. By (1), $\{(\lambda_\ell, S\varphi_\ell)\}$ is an eigendecomposition of T_{W_Δ} (up to ordering/sign), hence the graphon Eig-PE map for W_Δ can be chosen as

$$t^{\mathrm{eig}}_{W_\Delta}(u) = (S\varphi_1(u), \ldots, S\varphi_k(u)).$$

On the other hand, by Definition 3,

$$(St^{\mathrm{eig}}_G)(u) = (S\varphi_1(u), \ldots, S\varphi_k(u)).$$

Therefore $(St^{\mathrm{eig}}_G)(u) = t^{\mathrm{eig}}_{W_\Delta}(u)$ for all $u \in [0,1]$, provided the same ordering/sign convention is used.

(3) Projector-PE correspondence. Let $\Pi_k x = \sum_{\ell=1}^k \langle x, \varphi_\ell\rangle_n\,\varphi_\ell$ be the rank-k projector on $\mathbb{R}^n$ (with respect to $\langle\cdot,\cdot\rangle_n$), and let $\Pi_{W_\Delta,k}$ be the rank-k projector on $L^2([0,1])$ onto $\operatorname{span}\{S\varphi_1, \ldots, S\varphi_k\}$. Using L^2-orthonormality of $\{S\varphi_\ell\}_{\ell=1}^k$ and the identity

$$\langle Sx, S\varphi_\ell\rangle_{L^2} = \sum_{j=1}^n \int_{P_j} x_j\varphi_\ell(j)\,du = \frac{1}{n}x^\top\varphi_\ell = \langle x, \varphi_\ell\rangle_n,$$

we obtain, for any $x \in \mathbb{R}^n$,

$$\Pi_{W_\Delta,k}(Sx) = \sum_{\ell=1}^k \langle Sx, S\varphi_\ell\rangle_{L^2}\,S\varphi_\ell = \sum_{\ell=1}^k \langle x, \varphi_\ell\rangle_n\,S\varphi_\ell = S\left(\sum_{\ell=1}^k \langle x, \varphi_\ell\rangle_n\,\varphi_\ell\right) = S(\Pi_k x).$$

Take $R \in \mathbb{R}^{n\times m}$ with readout function $r := SR \in L^2([0,1];\mathbb{R}^m)$ channel-wise, so that $r_q = (SR)_q = S(R_{:q})$. Then for each q,

$$t^{\mathrm{proj}}_{W_\Delta,q} = \Pi_{W_\Delta,k} r_q = \Pi_{W_\Delta,k}(S(R_{:q})) = S(\Pi_k R_{:q}) = (St^{\mathrm{proj}}_G)_q.$$

hence $t^{\mathrm{proj}}_{W_\Delta}(u) = (St^{\mathrm{proj}}_G)(u)$ for all $u \in [0,1]$. This proves (3). $\square$

2 Stability of PE and Lipschitz network

Assumption 1 (For Eig-PE)**.** *Fix $k \in \mathbb{N}$. Let W be a graphon with bounded self-adjoint operator T_W. Let $\sigma(T_W)$ be the spectrum of T_W, and define* $\operatorname{dist}(a, S) := \inf_{s\in S} |a - s|$ *for $a \in \mathbb{R}$, $S \subseteq \mathbb{R}$. Assume the first k eigenvalues are simple and separated:*

$$\gamma_\ell := \operatorname{dist}\big(\lambda_\ell,\ \sigma(T_W) \setminus \{\lambda_\ell\}\big) \ \geq\ \gamma\ >\ 2\varepsilon\ >\ 0, \qquad \ell = 1, \ldots, k,$$

where ε is the operator perturbation level (e.g., $\|T_{W_{\pi(\Delta)}} - T_W\|_{L^2\to L^2} \leq \varepsilon$).

Assumption 2 (For Proj-PE)**.** *Fix $k, m \in \mathbb{N}$. Let W be a graphon with bounded self-adjoint operator T_W, and let $\sigma(T_W)$ denote its spectrum. Define* $\operatorname{dist}(a, S) := \inf_{s\in S} |a - s|$. *Assume the rank-$k$ spectral subspace is separated:*

$$\gamma_k := \min_{i\leq k}\ \operatorname{dist}\big(\lambda_i,\ \sigma(T_W) \setminus \{\lambda_1, \ldots, \lambda_k\}\big) \ \geq\ \gamma\ >\ 2\varepsilon\ >\ 0,$$

where ε denotes the operator perturbation level (e.g., $\|T_{W_{\pi(\Delta)}} - T_W\|_{L^2\to L^2} \leq \varepsilon$). Also assume the readout $r \in L^2([0, 1]; \mathbb{R}^m)$ used in $t^{\mathrm{proj}}_W = \Pi_{W,k} r$ satisfies

$$\|r\|_{L^2([0,1];\mathbb{R}^m)} \leq B.$$

Lemma 1 (Graph-to-Graphon Stability of PE)**.** *Fix $k \in \mathbb{N}$. Let W be a graphon with bounded self-adjoint operator T_W on $L^2([0, 1])$ and let Δ be the graph shift operator of a finite graph G sampled from W, and let $W_{\pi(\Delta)}$ be the induced step-graphon of $\pi(\Delta)$. Also $t_{\pi(G)}$ is defined from the eigenpairs of $\pi(\Delta)$. Assume*

$$\big\|T_{W_{\pi(\Delta)}} - T_W\big\|_{L^2\to L^2} \leq \varepsilon.$$

(a) ***Eig-PE stability.*** *Under Assumption 1, we have*

$$\big\|St^{\mathrm{eig}}_{\pi(G)} - t^{\mathrm{eig}}_W\big\|_{L^2([0,1];\mathbb{R}^k)} \ \leq\ C_{\mathrm{eig}}\,\varepsilon, \quad C_{\mathrm{eig}} := \sqrt{k}\max_{\ell\leq k} O(1/\gamma_\ell)$$

(b) ***Proj-PE stability.*** *Under Assumption 2, we have*

$$\big\|St^{\mathrm{proj}}_{\pi(G)} - t^{\mathrm{proj}}_W\big\|_{L^2([0,1];\mathbb{R}^m)} \ \leq\ C_{\mathrm{proj}}\,\varepsilon, \quad C_{\mathrm{proj}} := B\,O(1/\gamma_k).$$

Proof. (a) By Proposition 1 (Eig-PE correspondence), after fixing the same ordering/sign convention, we may choose the first k eigenfunctions of $T_{W_{\pi(\Delta)}}$ as $\{S\varphi_\ell\}_{\ell=1}^k$, where $\{(\lambda_\ell, \varphi_\ell)\}_{\ell=1}^k$ are eigenpairs of $\pi(\Delta)$. Hence

$$St^{\mathrm{eig}}_{\pi(G)} \ =\ t^{\mathrm{eig}}_{W_{\pi(\Delta)}}.$$

Under Assumption 1, the Davis–Kahan theorem for bounded self-adjoint operators yields eigenfunctions $\tilde{\phi}_\ell$ of $T_{W_{\pi(\Delta)}}$ (with a sign choice $\langle\tilde{\phi}_\ell, \phi_\ell\rangle_{L^2} \geq 0$) such that

$$\|\tilde{\phi}_\ell - \phi_\ell\|_{L^2([0,1])} \ \leq\ C_\ell\,\|T_{W_{\pi(\Delta)}} - T_W\|_{L^2\to L^2} \ \leq\ C_\ell\,\varepsilon, \qquad C_\ell = O(1/\gamma_\ell),\ \ell = 1, \ldots, k.$$

Taking $\tilde{\phi}_\ell := (t^{\mathrm{eig}}_{W_{\pi(\Delta)}})_\ell$ gives

$$\|t^{\mathrm{eig}}_{W_{\pi(\Delta)}} - t^{\mathrm{eig}}_W\|^2_{L^2([0,1];\mathbb{R}^k)} = \sum_{\ell=1}^k \|\tilde{\phi}_\ell - \phi_\ell\|^2_{L^2([0,1])} \ \leq\ k\Big(\max_{\ell\leq k} C_\ell\Big)^2\varepsilon^2.$$

Therefore,

$$\|St^{\mathrm{eig}}_{\pi(\Delta)} - t^{\mathrm{eig}}_{W}\|_{L^2([0,1];\mathbb{R}^k)} = \|t^{\mathrm{eig}}_{W_{\pi(\Delta)}} - t^{\mathrm{eig}}_{W}\|_{L^2([0,1];\mathbb{R}^k)} \;\le\; \sqrt{k}\Big(\max_{\ell\le k} C_\ell\Big)\varepsilon,$$

which proves (a) with $C_{\mathrm{eig}} := \sqrt{k}\max_{\ell\le k} C_\ell$.

(b) By Proposition 1 (Projector-PE correspondence),

$$St^{\mathrm{proj}}_{\pi(\Delta)} \;=\; t^{\mathrm{proj}}_{W_{\pi(\Delta)}}.$$

Under Assumption 2, the Davis–Kahan *sin*-Θ bound yields

$$\|\Pi_{W_{\pi(\Delta)},k} - \Pi_{W,k}\|_{L^2\to L^2} \;\le\; C\,\|T_{W_{\pi(\Delta)}} - T_W\|_{L^2\to L^2} \;\le\; C\,\varepsilon, \qquad C = O(1/\gamma_k).$$

Thus,

$$\|St^{\mathrm{proj}}_{\pi(\Delta)} - t^{\mathrm{proj}}_{W}\|_{L^2} = \|t^{\mathrm{proj}}_{W_{\pi(\Delta)}} - t^{\mathrm{proj}}_{W}\|_{L^2} = \|(\Pi_{W_{\pi(\Delta)},k} - \Pi_{W,k})r\|_{L^2} \le \|\Pi_{W_{\pi(\Delta)},k} - \Pi_{W,k}\|_{L^2\to L^2}\,\|r\|_{L^2} \le B\,C\,\varepsilon,$$

which proves (b) with $C_{\mathrm{proj}} := BC$. □

Corollary 1 (Graphon-to-graphon PE stability)**.** *Fix $k \in \mathbb{N}$. Let W, W' be two graphons with bounded self-adjoint operators $T_W, T_{W'}$ on $L^2([0,1])$. Assume there exists a measure-preserving bijection $\pi : [0,1] \to [0,1]$ such that*

$$\big\|T_{W^\pi} - T_{W'}\big\|_{L^2\to L^2} \le \varepsilon, \qquad W^\pi(u,v) := W(\pi(u),\pi(v)).$$

(a) ***Eig-PE stability.*** *Under Assumption 1 (applied to W' and W^π), we have*

$$\big\|t^{\mathrm{eig}}_{W^\pi} - t^{\mathrm{eig}}_{W'}\big\|_{L^2([0,1];\mathbb{R}^k)} \;\le\; C_{\mathrm{eig}}\,\varepsilon, \quad C_{\mathrm{eig}} := \sqrt{k}\max_{\ell\le k} O(1/\gamma_\ell).$$

(b) ***Proj-PE stability.*** *Under Assumption 2 (applied to W' and W^π), we have*

$$\big\|t^{\mathrm{proj}}_{W^\pi} - t^{\mathrm{proj}}_{W'}\big\|_{L^2([0,1];\mathbb{R}^m)} \;\le\; C_{\mathrm{proj}}\,\varepsilon, \quad C_{\mathrm{proj}} := B\,O(1/\gamma_k).$$

Proof. The claim follows by the same Davis–Kahan perturbation arguments as in Lemma 1, applied to the pair of bounded self-adjoint operators $(T_{W^\pi},\, T_{W'})$. □

Remark 1 (Eig-PE vs. Proj-PE). Eig-PE is a compact coordinate map $t^{\mathrm{eig}}_W(u) = (\phi_1(u), \dots, \phi_k(u))$, but its stability is gap-sensitive: small eigen-gaps γ_ℓ can induce large sign/rotation changes of eigenfunctions, making the coordinates unstable. Proj-PE instead encodes the *top-k spectral subspace* via the projector $t^{\mathrm{proj}}_W = \Pi_{W,k} r$, which is basis-invariant within $\mathrm{span}\{\phi_1, \dots, \phi_k\}$ and whose stability depends only on the subspace gap γ_k.

Practically, one may choose k by a stability criterion (avoid small gaps), and tune the readout dimension m and r/R to trade off cost and expressivity (random or structured/learnable readouts). When the spectrum is well separated, Eig-PE and Proj-PE become close (up to basis choice); when gaps are small, Proj-PE is typically more robust.

Assumption 3 (Lipschitz DeepSets)**.** *A DeepSets-type representation on a measure μ over $\mathbb{R}^k$ is of the form*

$$F_\theta(\mu) \;:=\; \rho_\theta\left(\int_{\mathbb{R}^k} \phi_\theta(z)\, d\mu(z)\right),$$

where $\phi_\theta : \mathbb{R}^k \to \mathbb{R}^d$ is the token embedding and $\rho_\theta : \mathbb{R}^d \to \mathbb{R}$ is the post-aggregation map. Assume ρ_θ is L_ρ-Lipschitz and and ϕ_θ is L_ϕ-Lipschitz.

In particular, for an empirical measure $\mu = \frac{1}{N}\sum_{i=1}^{N} \delta_{z_i}$,

$$F_\theta(\mu) \;=\; \rho_\theta\left(\frac{1}{N}\sum_{i=1}^{N} \phi_\theta(z_i)\right),$$

which recovers the standard DeepSets architecture.

Proposition 2. *Let F_θ satisfy Assumption 3, let μ and ν be probability measures on $\mathbb{R}^k$ with finite first moment. Then*

$$\big|F_\theta(\mu) - F_\theta(\nu)\big| \;\le\; L_\rho\, L_\phi\, \mathsf{W}_1(\mu, \nu),$$

where W_1 is the 1-Wasserstein distance on $\mathbb{R}^k$. In particular, if $\mathsf{W}_1(\mu,\nu) \le \varepsilon$, then $|F_\theta(\mu) - F_\theta(\nu)| \le L_\rho L_\phi\, \varepsilon$.

Proof. Let

$$m_\mu := \int_{\mathbb{R}^k} \phi_\theta(z)\, d\mu(z) \in \mathbb{R}^d, \qquad m_\nu := \int_{\mathbb{R}^k} \phi_\theta(z)\, d\nu(z) \in \mathbb{R}^d.$$

By L_ρ-Lipschitzness of ρ_θ,

$$|F_\theta(\mu) - F_\theta(\nu)| \le L_\rho \|m_\mu - m_\nu\|_2.$$

Fix any coupling $\pi \in \Pi(\mu, \nu)$ of $(Z, Z') \sim \pi$. Then

$$m_\mu - m_\nu = \int_{\mathbb{R}^k \times \mathbb{R}^k} \big(\phi_\theta(z) - \phi_\theta(z')\big)\, d\pi(z, z').$$

Taking norms and using Jensen (or the triangle inequality for Bochner integrals),

$$\|m_\mu - m_\nu\|_2 \le \int \|\phi_\theta(z) - \phi_\theta(z')\|_2\, d\pi(z, z') \le L_\phi \int \|z - z'\|_2\, d\pi(z, z').$$

Taking the infimum over all couplings π and using the primal definition of W_1,

$$\|m_\mu - m_\nu\|_2 \le L_\phi\, \mathsf{W}_1(\mu, \nu).$$

Combining the above inequalities gives

$$|F_\theta(\mu) - F_\theta(\nu)| \le L_\rho\, L_\phi\, \mathsf{W}_1(\mu, \nu),$$

as claimed. ☐

Assumption 4 (Lipschitz GNN)**.** *A graph neural network $\Phi := \Phi(H, M, \rho) : \mathbb{R}^k \to \mathbb{R}^d$ is* Lipschitz *if the following hold.*

(a) Contractive nonlinearity. $|\rho(x) - \rho(y)| \le |x - y|$ for all $x, y \in \mathbb{R}$.

(b) *Bounded operators. For all input graphs/graphons under consideration, the associated (self-adjoint) operators satisfy*

$$\|T_W\|_{L^2\to L^2} \le \Gamma \quad \text{(and likewise for induced step-graphons } W_\Delta\text{)}$$

(c) *Regular spectral filters. For every layer $\ell = 1,\ldots,L$ and channel pair (j,k), the filter $h_\ell^{jk} \in C^1[-\Gamma,\Gamma]$, $(h_\ell^{jk})'$ is Lipschitz on $[-\Gamma,\Gamma]$, and $\|h_\ell^{jk}\|_{L^\infty[-\Gamma,\Gamma]} \le 1$.*

(d) *Bounded mixing matrices. $\|M_\ell\|_\infty \le M$ for $\ell = 1,\ldots,L$ (in particular, $M=1$ yields depth-independent constants).*

Proposition 3 (Lipschitz stability of GNN outputs)**.** *Let $\Phi := \Phi(H,M,\rho)$ be an L-layer GNN satisfying Assumption 4. Let W_1, W_2 be graphons with bounded self-adjoint operators $T_{W_1}, T_{W_2} : L^2([0,1]) \to L^2([0,1])$. Let $x_1, x_2 \in L^2([0,1];\mathbb{R}^k)$ be input feature maps satisfying $\|x_1\|_{L^2([0,1];\mathbb{R}^k)} \le 1$ and $\|x_2\|_{L^2([0,1];\mathbb{R}^k)} \le 1$. Then there exists a constant $C_\Phi > 0$, depending only on Φ, such that*

$$\big\|\Phi_{W_1}(x_1) - \Phi_{W_2}(x_2)\big\|_{L^2([0,1];\mathbb{R}^d)} \;\le\; C_\Phi\Big(\|T_{W_1} - T_{W_2}\|_{L^2\to L^2} + \|x_1 - x_2\|_{L^2([0,1];\mathbb{R}^k)}\Big).$$

Proof. We follow the proof of [4, Lemma 5.16] (Appendix A.8) and adapt it to accommodate arbitrary discrepancies between the two inputs/graphons.

Step 0: layerwise notation. Let $F_0 = k$ and $F_L = d$. For $i \in \{1,2\}$, define the layerwise feature maps $z_i^{(0)} := x_i \in L^2([0,1];\mathbb{R}^{F_0})$ and for $\ell = 1,\ldots,L$,

$$z_i^{(\ell),j} := \rho\left(\sum_{k=1}^{F_{\ell-1}} m_\ell^{jk}\, h_\ell^{jk}(T_{W_i})\, z_i^{(\ell-1),k}\right), \qquad j = 1,\ldots,F_\ell,$$

where $M_\ell = (m_\ell^{jk})_{j,k}$ and ρ is contractive (1-Lipschitz).

Let $M := \max_\ell \|M_\ell\|_\infty$. By Assumption 4(b), $\|T_{W_i}\|_{2\to2} \le \Gamma$. Moreover, since each filter h satisfies $h \in C^1([-\Gamma,\Gamma])$ with Lipschitz continuous derivative, the functional calculus bound (cf. [4, Eq.(22)]) yields that there exists a constant $C_h > 0$ such that for all self-adjoint operators A, B with spectrum in $[-\Gamma,\Gamma]$,

$$\|h(A) - h(B)\|_{2\to2} \le C_h \|A - B\|_{2\to2}.$$

Let $C := \max_{\ell,j,k} C_{h_\ell^{jk}}$.

Step 1: uniform bound on feature norms. Using $\|\rho\|_{\mathrm{Lip}} \le 1$, $\|h_\ell^{jk}(T_{W_i})\|_{2\to2} \le \|h_\ell^{jk}\|_\infty \le 1$, and $\|M_\ell\|_\infty \le M$, we have for each ℓ,

$$\max_{j\le F_\ell} \|z_i^{(\ell),j}\|_{L^2} \le M \max_{k\le F_{\ell-1}} \|z_i^{(\ell-1),k}\|_{L^2} \le M^\ell \|x_i\|_{L^2}.$$

Under the normalization $\|x_i\|_{L^2} \le 1$, this gives $\max_{j\le F_\ell} \|z_i^{(\ell),j}\|_{L^2} \le M^\ell$ for $i = 1,2$.

Step 2: perturbation recursion. Define $R_\ell := \max_{j\le F_\ell} \|z_1^{(\ell),j} - z_2^{(\ell),j}\|_{L^2}$. By contractivity of ρ and triangle inequality, for each ℓ,

$$\begin{aligned}
R_\ell &\le \max_j \left\| \sum_k m_\ell^{jk}\Big(h_\ell^{jk}(T_{W_1}) z_1^{(\ell-1),k} - h_\ell^{jk}(T_{W_2}) z_2^{(\ell-1),k}\Big)\right\|_{L^2} \\
&\le M \max_k \|h_\ell^{jk}(T_{W_1}) z_1^{(\ell-1),k} - h_\ell^{jk}(T_{W_2}) z_2^{(\ell-1),k}\|_{L^2} \\
&\le M \max_k \Big(\|(h_\ell^{jk}(T_{W_1}) - h_\ell^{jk}(T_{W_2})) z_1^{(\ell-1),k}\|_{L^2} + \|h_\ell^{jk}(T_{W_2})(z_1^{(\ell-1),k} - z_2^{(\ell-1),k})\|_{L^2}\Big) \\
&\le M\Big(C\|T_{W_1} - T_{W_2}\|_{2\to2} \cdot \max_k \|z_1^{(\ell-1),k}\|_{L^2} + R_{\ell-1}\Big).
\end{aligned}$$

Using Step 1, $\max_k \|z_1^{(\ell-1),k}\|_{L^2} \leq M^{\ell-1}$, we obtain

$$R_\ell \leq MR_{\ell-1} + CM^\ell \|T_{W_1} - T_{W_2}\|_{2\to 2}.$$

Unrolling this recursion and noting $R_0 = \|x_1 - x_2\|_{L^2}$ yields

$$R_L \leq M^L \|x_1 - x_2\|_{L^2} + (CL)M^L \|T_{W_1} - T_{W_2}\|_{2\to 2}.$$

Step 3: conclude the L^2 output bound. Finally,

$$\|\Phi_{W_1}(x_1) - \Phi_{W_2}(x_2)\|_{L^2([0,1];\mathbb{R}^{F_L})} \leq \sqrt{F_L}\, R_L \leq C_\Phi \Big(\|T_{W_1} - T_{W_2}\|_{2\to 2} + \|x_1 - x_2\|_{L^2} \Big),$$

where $C_\Phi := \sqrt{F_L}\, M^L \max\{1, CL\}$ depends only on the architecture/filter family in Φ. □

3 Main Results

Theorem 1 (Graph-to-Graphon Stability of NNs with PE input)**.** *Assume $\|T_{W_{\pi(\Delta)}} - T_W\|_{L^2\to L^2} \leq \varepsilon$ for some permutation π.*

(a) ***DeepSets with PE-token measures.*** *Let F_θ satisfy Assumption 3. Let $t_G : [n] \to \mathbb{R}^k$ be a graph PE token map and $t_W : [0,1] \to \mathbb{R}^k$ a graphon PE map, and define*

$$\mu(t_G) := (S\pi t_G)_\# \lambda, \qquad \mu(t_W) := (t_W)_\# \lambda,$$

where λ is the uniform measure on $[0,1]$ and $S\pi t_G$ is the induced step map. Assume the corresponding PE stability bound holds as constructed in Lemma 1:

$$\|S\pi t_G - t_W\|_{L^2([0,1];\mathbb{R}^k)} \leq C_{\mathrm{PE}}\varepsilon.$$

Then

$$\big|F_\theta(\mu(t_G)) - F_\theta(\mu(t_W))\big| \;\leq\; L_\rho L_\phi\, C_{\mathrm{PE}}\, \varepsilon.$$

(b) ***GNN with induced step PE signals.*** *Let Φ satisfy Assumption 4. Assume the corresponding PE stability bound holds as constructed in Lemma 1:*

$$\|S\pi t_G - t_W\|_{L^2([0,1];\mathbb{R}^k)} \leq C_{\mathrm{PE}}\varepsilon.$$

Then there exists a constant $C > 0$ depending only on Φ and C_{PE} such that

$$\big\|\Phi_{W_{\pi(\Delta)}}(S\pi t_G) - \Phi_W(t_W)\big\|_{L^2([0,1];\mathbb{R}^d)} \;\leq\; C\,\varepsilon.$$

Proof of (a). Let λ be the uniform probability measure on $[0,1]$. Consider the coupling Γ between $\mu(t_G)$ and $\mu(t_W)$ defined as the law of

$$\big(S\pi t_G(U),\, t_W(U)\big) \quad \text{with } U \sim \lambda.$$

Then $\Gamma \in \Pi(\mu(t_G), \mu(t_W))$, hence by the definition of W_1,

$$\begin{aligned} W_1\big(\mu(t_G), \mu(t_W)\big) &\leq \int_{[0,1]} \big\|S\pi t_G(u) - t_W(u)\big\|_2 \, d\lambda(u) \\ &= \big\|S\pi t_G - t_W\big\|_{L^1([0,1];\mathbb{R}^k)} \;\leq\; \big\|S\pi t_G - t_W\big\|_{L^2([0,1];\mathbb{R}^k)}. \end{aligned}$$

By the assumed PE stability bound, $\|S\pi t_G - t_W\|_{L^2([0,1];\mathbb{R}^k)} \le C_{\mathrm{PE}}\varepsilon$, so

$$W_1\big(\mu(t_G), \mu(t_W)\big) \le C_{\mathrm{PE}}\varepsilon.$$

Finally, applying Proposition 2 (DeepSets is $L_\rho L_\phi$-Lipschitz w.r.t. W_1) yields

$$\big|F_\theta(\mu(t_G)) - F_\theta(\mu(t_W))\big| \ \le\ L_\rho L_\phi\, W_1\big(\mu(t_G), \mu(t_W)\big) \ \le\ L_\rho L_\phi\, C_{\mathrm{PE}}\,\varepsilon.$$

□

Proof of (b). Apply Proposition 3 to the pair

$$(W_1, x_1) = (W_{\pi(\Delta)},\, S\pi t_G), \qquad (W_2, x_2) = (W,\, t_W).$$

Then

$$\|\Phi_{W_{\pi(\Delta)}}(S\pi t_G) - \Phi_W(t_W)\|_{L^2([0,1];\mathbb{R}^d)} \le C_\Phi\Big(\|T_{W_{\pi(\Delta)}} - T_W\|_{L^2\to L^2} + \|S\pi t_G - t_W\|_{L^2([0,1];\mathbb{R}^k)}\Big).$$

Using $\|T_{W_{\pi(\Delta)}} - T_W\|_{L^2\to L^2} \le \varepsilon$ and the assumed PE stability bound $\|S\pi t_G - t_W\|_{L^2([0,1];\mathbb{R}^k)} \le C_{\mathrm{PE}}\varepsilon$, we obtain

$$\|\Phi_{W_{\pi(\Delta)}}(S\pi t_G) - \Phi_W(t_W)\|_{L^2([0,1];\mathbb{R}^d)} \le C_\Phi(1 + C_{\mathrm{PE}})\,\varepsilon.$$

Thus the claim holds with $C := C_\Phi(1 + C_{\mathrm{PE}})$. □

Remark 2 (Choice of PE). C_{PE} denotes the PE-token stability constant used in Theorem 1, i.e., $\|S\pi t_G - t_W\|_{L^2} \le C_{\mathrm{PE}}\|T_{W_{\pi(\Delta)}} - T_W\|_{L^2\to L^2}$. We instantiate it by the corresponding PE bound: for Eig-PE take $C_{\mathrm{PE}} := C_{\mathrm{eig}}$ from Lemma 1(a), and for Proj-PE take $C_{\mathrm{PE}} := C_{\mathrm{proj}}$ from Lemma 1(b). For notational convenience we use the same k to denote the PE output dimension for both variants.

Corollary 2 (Graphon-to-graphon stability of NNs with PE input)**.** *Let W, W' be two (bounded, symmetric) graphons with bounded self-adjoint operators $T_W, T_{W'}$ on $L^2([0,1])$. Assume there exists a measure-preserving bijection $\pi : [0,1] \to [0,1]$ such that*

$$\big\|T_{W^\pi} - T_{W'}\big\|_{L^2\to L^2} \le \varepsilon, \qquad W^\pi(u,v) := W(\pi(u), \pi(v)).$$

Assume the corresponding PE stability bound holds as constructed in Corollary 1:

$$\|t_{W^\pi} - t_{W'}\|_{L^2([0,1];\mathbb{R}^k)} \le C_{\mathrm{PE}}\,\varepsilon.$$

Then:

(a) ***DeepSets on PE-token measures.*** *Let F_θ satisfy Assumption 3, and define*

$$\mu(t_{W^\pi}) := (t_{W^\pi})_\#\lambda, \qquad \mu(t_{W'}) := (t_{W'})_\#\lambda,$$

where λ is the uniform probability measure on $[0,1]$. Then

$$\big|F_\theta(\mu(t_{W^\pi})) - F_\theta(\mu(t_{W'}))\big| \ \le\ L_\rho L_\phi\, C_{\mathrm{PE}}\,\varepsilon.$$

(b) ***GNN on graphons with PE signals.*** *Let Φ satisfy Assumption 4. Then*

$$\big\|\Phi_{W^\pi}(t_{W^\pi}) - \Phi_{W'}(t_{W'})\big\|_{L^2([0,1];\mathbb{R}^d)} \ \le\ C_\Phi\,(1 + C_{\mathrm{PE}})\,\varepsilon.$$

Theorem 2 (Graph-to-graph stability via a graphon error decomposition)**.** *Let $\Delta^{(1)} \in \mathbb{R}^{n_1 \times n_1}$ and $\Delta^{(2)} \in \mathbb{R}^{n_2 \times n_2}$ be two finite graph operators, sampled (possibly with different sizes) from two underlying graphons $W^{(1)}$ and $W^{(2)}$, respectively. Let π_1, π_2 be node permutations and consider the induced step-graphons $W_{\pi_1(\Delta^{(1)})}$ and $W_{\pi_2(\Delta^{(2)})}$ with operators $T_{W_{\pi_1(\Delta^{(1)})}}$ and $T_{W_{\pi_2(\Delta^{(2)})}}$.*

Assume the following three discrepancies are bounded:

$$\varepsilon_1 := \left\|T_{W_{\pi_1(\Delta^{(1)})}} - T_{W^{(1)}}\right\|_{L^2 \to L^2}, \qquad \varepsilon_2 := \left\|T_{W_{\pi_2(\Delta^{(2)})}} - T_{W^{(2)}}\right\|_{L^2 \to L^2},$$

and there exists a measure-preserving bijection $\pi : [0,1] \to [0,1]$ such that

$$\varepsilon_{\text{gra}} := \left\|T_{(W^{(1)})^\pi} - T_{W^{(2)}}\right\|_{L^2 \to L^2}.$$

Let $t_{G^{(1)}}, t_{G^{(2)}}$ be the (discrete) graph PE token maps, and $t_{W^{(1)}}, t_{W^{(2)}}$ the corresponding graphon PE maps (same PE type throughout). Assume the PE stability bounds in Lemma 1 hold with constants $C_{\text{PE}}^{(1)}$ and $C_{\text{PE}}^{(2)}$, and define

$$C_{\text{PE}} \;:=\; \max\left\{C_{\text{PE}}^{(1)},\, C_{\text{PE}}^{(2)}\right\}.$$

Namely,

$$\|S\pi_1 t_{G^{(1)}} - t_{W^{(1)}}\|_{L^2} \le C_{\text{PE}}\,\varepsilon_1, \qquad \|S\pi_2 t_{G^{(2)}} - t_{W^{(2)}}\|_{L^2} \le C_{\text{PE}}\,\varepsilon_2,$$

and

$$\|t_{(W^{(1)})^\pi} - t_{W^{(2)}}\|_{L^2} \le C_{\text{PE}}\,\varepsilon_{\text{gra}}.$$

Then:

(a) ***DeepSets discrepancy decomposition.*** *Let F_θ satisfy Assumption 3, and define*

$$\mu^{(1)} := (S\pi_1 t_{G^{(1)}})_{\#}\lambda, \qquad \mu^{(2)} := (S\pi_2 t_{G^{(2)}})_{\#}\lambda.$$

Then

$$\left|F_\theta(\mu^{(1)}) - F_\theta(\mu^{(2)})\right| \;\le\; L_\rho L_\phi\, C_{\text{PE}}\left(\varepsilon_1 + \varepsilon_{\text{gra}} + \varepsilon_2\right).$$

(b) ***GNN discrepancy decomposition.*** *Let Φ satisfy Assumption 4. Then*

$$\left\|\Phi_{W_{\pi_1(\Delta^{(1)})}}\big(S\pi_1 t_{G^{(1)}}\big) \circ \pi - \Phi_{W_{\pi_2(\Delta^{(2)})}}\big(S\pi_2 t_{G^{(2)}}\big)\right\|_{L^2([0,1];\mathbb{R}^d)} \;\le\; C_\Phi\,(1 + C_{\text{PE}})\left(\varepsilon_1 + \varepsilon_{\text{gra}} + \varepsilon_2\right).$$

Proof. Both parts follow from a triangle inequality with two applications of Theorem 1 (graph-to-graphon) and one application of Corollary 2 (graphon-to-graphon), using the three discrepancies $\varepsilon_1, \varepsilon_{\text{gra}}, \varepsilon_2$. □

Remark 3 (DeepSets vs. GNN)*.* A GNN Φ can take *extra node features* in addition to PE (not only spectral/PE features) and supports both node-level prediction and graph-level prediction (via pooling/readout). DeepSets F_θ is simpler: it acts on the PE-token measure directly, giving a permutation-invariant graph embedding with lower architectural/compute complexity, which is often sufficient for pure graph classification.

4 Practical Guidelines: Stability–Expressivity and Data Curation

4.1 Choosing PE & hyperparameters

Our analysis suggests that the dominant stability bottleneck of spectral PEs is the *conditioning* of the spectrum, i.e., eigengaps. In particular, eigenvector-based PEs are sensitive to small gaps (and basis/sign ambiguities), while projector-based PEs are inherently basis-invariant and inherit subspace stability, making them preferable under domain/size shifts.

Relatedly, *RFP* [2] avoids explicit eigendecomposition and instead performs a subspace-iteration–style procedure: starting from random node features, it alternates propagation by a graph operator and normalization/orthogonalization across P steps, and uses the whole trajectory as PE. In the idealized limit, this process recovers the *dominant* k-dimensional eigenspace of the propagation operator (i.e., the invariant subspace associated with the largest-magnitude eigenvalues; for PSD diffusion operators, this coincides with the top-k largest eigenvalues), which aligns it conceptually with projector/probe-based PEs.

Beyond the hard projector, *SPE* [3] can be viewed as a more expressive projector-style variant: rather than using only the hard top-k projector $U_k U_k^\top$, it inserts an *eigenvalue-equivariant* (basis-consistent) transform in the spectral domain. Abstractly, a broad class of such spectral-domain constructions can be written as

$$t^{\mathrm{spe}}(W) \;\approx\; U\,G(\Lambda)\,U^\top r,$$

for an eigenvalue-dependent map G and a probe/readout r (e.g., pooling can be interpreted as choosing a fixed readout such as an all-ones vector). This eigenvalue-aware mixing retains finer information about the spectral distribution (e.g., soft frequency partitioning) beyond the mere top-k subspace, and can thus improve expressivity in regimes where different graphons share similar leading subspaces but differ in eigenvalue profiles. From a stability perspective, this also clarifies the key tradeoff: proj-PE stability is governed by a *subspace condition number* (via a gap γ_k), whereas SPE-style stability is additionally controlled by the *smoothness* of the eigenvalue-dependent transform (sharp/step-like gating improves expressivity but may reintroduce instability reminiscent of hard cutoffs; smoother gating improves robustness under shifts).

Practically, we recommend selecting PE type and dimensions by explicitly budgeting the perturbation amplification:

$$\text{(stability)}\qquad \|t_{W_1} - t_{W_2}\| \le C_{\mathrm{PE}}(k,r)\,\varepsilon, \qquad C^{\mathrm{eig}}_{\mathrm{PE}}(k) \propto \sqrt{k}\max_{\ell\le k}\gamma_\ell^{-1}, \quad C^{\mathrm{proj}}_{\mathrm{PE}}(k,r) \propto \|r\|_{L^2}\gamma_k^{-1},$$

where ε denotes the operator discrepancy (graph-to-graphon or graphon-to-graphon) and γ_ℓ, γ_k are the relevant eigengaps. This motivates: (i) use **proj-PE/SPE** as the default choice when gaps are not well separated or multiplicities are common; (ii) choose k as the largest value such that the gap condition remains stable (e.g., ε/γ_k below a target threshold), since increasing k improves expressivity but can degrade stability when gaps shrink; (iii) choose the probe/readout dimension m large enough to avoid information collapse, with *normalized probes* so that $\|r\|_{L^2}$ remains $O(1)$ as m grows; and (iv) treat the downstream embedding size m as an expressivity knob, while controlling the Lipschitz constant of the invariant backbone (pooled GNN/DeepSets) to prevent the stability constant from scaling unfavorably with width.

Takeaway. Proj-PE offers clean, basis-invariant guarantees when a stable gap criterion is satisfied (moderate k with non-negligible γ_k). SPE provides a more expressive projector-style alternative via eigenvalue-aware soft mixing, and can be advantageous when eigengaps are small or spectra are clustered, since it can preserve expressivity while mitigating the instability caused by hard eigenspace partitioning.

4.2 Experimental protocol

We consider a controlled setting with a finite collection of graphons $\{W_c\}_{c=1}^C$ and sample graphs $G \sim W_c$ with the graphon index c as the supervised label. To avoid explicit node alignment, the predictor uses a permutation-invariant architecture (DeepSets or pooled GNN), with spectral PE tokens as the initial node features. Our experiments target two complementary claims.

Data curation: fewer large graphs are more valuable than many small graphs under a fixed sampling budget. We compare training sets under fixed budgets such as total nodes $\sum_i n_i$ or total edge-sampling pairs $\sum_i n_i^2$, contrasting *many small* graphs versus *few large* graphs from the same graphon mixture. We expect fewer large graphs to yield smaller graph-to-graphon discrepancy (hence more stable PE tokens) and better test performance on larger graphs, since large graphs more closely approximate the underlying graphon and cover richer structural patterns. Motivated by this, we further develop a **graphon-guided merging** pre-processing: estimate a step-graphon (or edge-probability matrix) $\widehat{W}$ from multiple small graphs, use $\widehat{W}$ to guide the construction of a larger synthetic graph (by stitching/merging subgraphs in the latent coordinate or probability space), and then compute PEs on the denoised/merged object. We expect this preprocessing to reduce token discrepancy and narrow train–test error under size shifts.

Stability diagnostics: PE type and (k, m, r) help predict generalization under shifts. Overall, we advocate reporting both (a) *token discrepancy* under perturbations (e.g., $\|t_{G_1} - t_{G_2}\|$ for graphs sampled from the same graphon at different sizes) and (b) *task loss/error*, as a direct diagnostic of the stability–expressivity tradeoff induced by PE choices and hyperparameters.

We sweep PE choices (eig-PE vs proj-PE/SPE) and hyperparameters (k, m, r), and evaluate (i) token stability metrics (e.g., within-graphon discrepancies across sizes/domains) and (ii) downstream test loss/error. We expect monotone trends consistent with theory: increasing k improves expressivity but can hurt stability when eigengaps shrink; proj-PE/SPE alleviates this sensitivity; and appropriately normalized probes allow increasing r to improve information retention without inflating instability. These experiments provide actionable guidance on selecting PE and hyperparameters to balance stability and expressivity for domain- and size-robust learning.

4.3 Experimental plan

Goal. We instantiate Sec. 4.2 as a controlled graphon-classification benchmark to validate: (i) *budgeted data curation*: under a fixed sampling budget, *few large* graphs outperform *many small* graphs for size-shift generalization; and (ii) *stability diagnostics*: PE type and hyperparameters predict the stability–expressivity tradeoff under size/domain shifts.

Data collection (graphon construction + graph sampling + budget control). We fix a collection of C graphons $\{W_c\}_{c=1}^C$, e.g. Fourier graphons. For the Fourier family, we use a circulant kernel

$$W(x, y) \;=\; \rho + \sum_{m=1}^{M} a_m \cos\bigl(2\pi m(x-y)\bigr), \qquad \sum_m |a_m| \le \min(\rho, 1-\rho),$$

which has matched edge density ρ and constant degree function $\int_0^1 W(x, y)\, dy = \rho$, while varying the spectral profile via $\{a_m\}$. We sample graphs $G \sim W_c$ via latent coordinates $u_1, \dots, u_n \overset{\text{iid}}{\sim} \mathrm{Unif}[0, 1]$ and edges $A_{ij} \sim \mathrm{Bernoulli}(W_c(u_i, u_j))$ for $1 \le i < j \le n$.

We impose a size shift by evaluating on a target test-size support $\mathcal{N}_{\text{te}} = \{64, 128, 256, 512, 768, 1024\}$, while training is restricted to $\mathcal{N}_{\text{tr}} := \{64, 128, 256, 512\}$ (i.e., no access to 768/1024 during training). In addition to size shift, we optionally study a *graphon shift* where the test set contains *unseen* graphons $\{W_c^{\text{te}}\}$ that are not used in training, e.g., by perturbing Fourier coefficients.

To study budgeted curation with a single, interpretable fairness knob, we use the *node budget* $B_V := \sum_{i=1}^{M} n_i$. Under a fixed B_V (e.g., $B_V \in \{10^5, 2 \cdot 10^5\}$), we compare a *size-allocation path* parameterized by $\lambda \in [0, 1]$ that interpolates between *many-small* and *few-large* while keeping the *training support set* fixed to $\mathcal{N}_{\text{tr}}$: for each increment of λ, we replace a small block of graphs at sizes $\{64, 128\}$ with a single larger graph at sizes $\{256, 512\}$, chosen so that the total node budget remains unchanged (e.g., replacing four 64/128 graphs by one 256/512 graph, depending on the exact mixture in the block). The endpoint $\lambda = 0$ corresponds to *many-small* (budget concentrated on $\{64, 128\}$) and $\lambda = 1$ corresponds to *few-large* (budget concentrated on $\{256, 512\}$). Finally, we consider a merging setting, where we only observe small training graphs but augment with synthetic large graphs by first estimating a graphon $\widehat{W}$ using the SAS (sorting-and-smoothing) estimator [1]: we start with a label-guided version that fits $\widehat{W}_c$ using *only* the training graphs from class c (no test graphs).

Models and PE tokens. We use permutation-invariant **DeepSets** to avoid explicit node alignment: a node MLP followed by pooling and an MLP head. Initial node features are spectral PE tokens, comparing **eig-PE** and **proj-PE/SPE**, with hyperparameters (k, m, r). To benchmark, we include four strong baselines: (i) **Degree-hist + MLP**, which feeds a binned degree histogram (optionally concatenated with basic graph statistics such as edge density) into a shallow MLP; (ii) a **graph kernel** baseline (e.g., Weisfeiler–Lehman subtree kernel) with an SVM classifier; (iii) **Graph2Vec + MLP**; and (iv) a pooled **GNN** baseline (e.g., GIN with sum pooling) trained on the same graphs. Unless otherwise specified, we tune each baseline with the same validation protocol and report its best validation-selected performance.

Evaluation metrics. We report three metrics:

(i) **Test error.** Graphon-classification error on test graphs $G \sim W_c^{\text{te}}$ with $n \in \mathcal{N}_{\text{te}}$.

(ii) **Train–test PE discrepancy.** For each graph G, let $\{t_i^{(G)}\}_{i \in V(G)} \subset \mathbb{R}^k$ denote its node tokens and define the *per-graph* empirical token measure

$$\mu_G \;:=\; \frac{1}{|V(G)|} \sum_{i \in V(G)} \delta_{t_i^{(G)}}.$$

Given a dataset $\mathcal{S}$ of graphs, we define the *graph-uniform* aggregate token measure

$$\mu_{\mathcal{S}} \;:=\; \frac{1}{|\mathcal{S}|} \sum_{G \in \mathcal{S}} \mu_G.$$

We then define the dataset-level shift as

$$\text{Discrepancy}_{\text{set}}(\mathcal{S}_{\text{tr}}, \mathcal{S}_{\text{te}}) \;:=\; \mathsf{W}_1\big(\mu_{\mathcal{S}_{\text{tr}}},\, \mu_{\mathcal{S}_{\text{te}}}\big).$$

Computation. In practice, we estimate each μ_G by uniformly subsampling m node tokens from $V(G)$ (without replacement), so that each graph contributes the same number of token samples. We approximate W_1 using the sliced Wasserstein distance with R random one-dimensional projections.

(iii) **Eigengap statistics.** We compute eigengap proxies such as $\min_{\ell \le k} \gamma_\ell(G)$ (or $\gamma_k(G)$) to diagnose spectral conditioning and relate it to instability and downstream error.

Experiment 1: Size matters and merging works.

- **Setup.** Under each fixed budget (node or pair budget), we compare three training pipelines: (a) *many-small*, (b) *few-large*, and (c) *many-small+merging*. For merging, we estimate a step-graphon (or edge-probability matrix) $\widehat{W}$ from many small graphs (e.g., histogram/step or smoothing), synthesize a larger graph $\widetilde{G}$ by sampling edges from Bernoulli($\widehat{W}(\tilde{u}_i, \tilde{u}_j)$), and include $\widetilde{G}$ as an augmented training example.

- **Reporting.** For each test size $n \in \mathcal{N}_{\mathrm{te}}$, we report test error, and we additionally report $\mathrm{Discrepancy}_{\mathrm{set}}$ between training-size and test-size graphs within each class, together with eigengap statistics.

- **Expected results.** We expect (i) *few-large* to improve large-n test performance and reduce train–test $\mathrm{Discrepancy}_{\mathrm{set}}$ relative to many-small; and (ii) *merging* to narrow this gap by reducing token discrepancy (and improving conditioning) for the many-small regime, yielding consistent gains on large-n tests.

Experiment 2: Hyperparameters predict stability–expressivity trends.

- **Setup.** We sweep PE choices and hyperparameters, e.g., $k \in \{8, 16, 32, 64, 128\}$ for eig-PE, and (k, m, r) grids for proj-PE/SPE.

- **Reporting.** We plot (a) test error versus k (or m, r), (b) train–test $\mathrm{Discrepancy}_{\mathrm{set}}$ versus hyperparameters, and (c) stability–performance scatter plots (test error vs. $\mathrm{Discrepancy}_{\mathrm{set}}$), annotated by eigengap statistics.

- **Expected results.** We expect monotone tradeoffs consistent with theory: increasing k improves expressivity but can worsen stability when eigengaps shrink (raising $\mathrm{Discrepancy}_{\mathrm{set}}$ and test error under shift); proj-PE/SPE alleviates eigengap sensitivity; and increasing probe dimension r can improve information retention without inflating instability when probes are properly normalized.

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